

Proof. **Trivial (degenerate) cases.**

1. **Case** $n = 0$.

$$(1 + x)^0 = 1 \quad \text{and} \quad 1 + 0 \cdot x = 1.$$

Equality holds.

2. **Case** $n = 1$.

$$(1 + x)^1 = 1 + x = 1 + 1 \cdot x.$$

Equality holds.

3. **Case** $x = 0$. For any $n \geq 0$,

$$(1 + 0)^n = 1 = 1 + n \cdot 0.$$

Equality holds.

These three situations constitute all edge cases that will be excluded from the induction argument that follows.

Base of induction ($n = 2$). Using the binomial theorem (or direct multiplication) we have the algebraic identity

$$(1 + x)^2 = 1 + 2x + x^2,$$

and since $x^2 \geq 0$ for every real x ,

$$(1 + x)^2 = 1 + 2x + x^2 \geq 1 + 2x.$$

Thus the inequality holds for $n = 2$.

Inductive step.

Inductive hypothesis. Assume that for some integer $k \geq 2$ and for all $x \geq -1$,

$$(1 + x)^k \geq 1 + kx. \quad (\text{IH})$$

Multiplication by a non-negative factor. Because $x \geq -1$ we have $1 + x \geq 0$. Multiplying both sides of (??) by $1 + x$ preserves the inequality:

$$\begin{aligned} (1 + x)^{k+1} &= (1 + x)^k (1 + x) \\ &\geq (1 + kx)(1 + x). \end{aligned} \quad (1)$$

Expansion of the right-hand side. The product expands as

$$(1 + kx)(1 + x) = 1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x. \quad (2)$$

Since $k \geq 0$ and $x^2 \geq 0$, the term kx^2 is non-negative, giving the last inequality.

Combining (??) and (??) yields

$$(1 + x)^{k+1} \geq 1 + (k + 1)x.$$

Thus the statement holds for $k + 1$. By mathematical induction the inequality is true for every integer $n \geq 2$.

Together with the trivial cases handled in Section 1, the inequality holds for **all** integers $n \geq 0$ and all real numbers $x \geq -1$.

Equality conditions. Equality in the inductive step requires simultaneously

1. equality in the hypothesis $(1 + x)^k = 1 + kx$; and
2. vanishing of the non-negative term kx^2 .

Because $k \geq 0$, the second condition forces either $k = 0$ (i.e. $n = 0$) or $x = 0$. The case $n = 1$ gives equality for every admissible x because the inequality reduces to the identity $1 + x = 1 + x$. The boundary point $x = -1$ yields $(1 - 1)^n = 0$ and $1 - n$; equality occurs only when $n = 1$, already included.

Hence **equality holds precisely when**

$$n = 0 \quad \text{or} \quad n = 1 \quad \text{or} \quad x = 0.$$

Conclusion. For every real number $x \geq -1$ and every integer $n \geq 0$,

$$(1 + x)^n \geq 1 + nx$$

with equality exactly in the cases $n = 0$, $n = 1$ or $x = 0$. □ □